

Data-Driven Dashboards for Enhancing Investor Reporting Transparency in American Capital Markets

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Abstract

The rapid digitalization of financial markets has transformed the way firms communicate information to investors. This study investigates whether data-driven dashboards enhance investor reporting transparency and influence market valuation in U.S. capital markets. Using secondary data from 47 S&P 500 companies across eight sectors, this research applies descriptive statistics, correlation analysis, and multiple regression models to evaluate the relationships among digital reporting quality, ESG performance, and market outcomes, as measured by price-to-earnings ratios. The results indicate that companies implementing advanced dashboard-based reporting systems receive significantly higher market valuation premiums. The quality of digital reporting systems shows a strong positive relationship with price-to-earnings ratios ($\beta = 402.97$, $p < 0.01$). Furthermore, the interaction analysis reveals that the valuation impact of ESG performance varies depending on the sophistication of digital transparency infrastructure. Sectoral results show that technology and communication service firms have the most advanced reporting dashboards, while the energy and industrial sectors have stronger ESG scores. These findings highlight the strategic role of digital transparency infrastructure in reducing information asymmetry and improving investor communication. The study contributes to digital transformation research by demonstrating that investor-oriented reporting technologies can generate measurable financial value in capital markets.

Keywords: capital markets, digital transformation, ESG performance, investor reporting, transparency dashboards

1. Introduction

The digital transformation of financial markets has significantly reshaped how organizations communicate information to investors and other stakeholders. Traditional financial reporting, which was historically dominated by static annual and quarterly reports, is increasingly being supplemented or replaced by digital platforms that enable real-time and dynamic disclosure of corporate information (Ma et al., 2025). This transition reflects a broader global trend in which organizations invest heavily

in digital technologies to improve transparency, efficiency, and competitiveness in capital markets (Shinozaki et al., 2025).

One of the most affected areas is investor communication. Technological advancements such as Environmental, Social, and Governance (ESG) integration, artificial intelligence–assisted reporting, and data-driven dashboards are increasingly adopted to enhance the accessibility and usability of financial information. These technologies not only improve operational efficiency but also transform the way companies interact with investors by enabling faster, more comprehensive, and more interactive information sharing (Prihandini & Sari, 2025). In modern financial markets, institutional and individual investors demand timely, detailed information to support rapid decision-making, particularly in volatile market environments and amid growing ESG awareness (Fianko et al., 2025). Consequently, organizations are increasingly adopting digital reporting technologies to strengthen transparency, credibility, and investor trust.

Traditional reporting mechanisms remain essential for regulatory compliance; however, they often struggle to meet the increasing demand for immediacy, interactivity, and contextual data analysis. Static reporting formats may limit investors' ability to access and interpret complex financial information efficiently (Rydzewski, 2025). In contrast, digital dashboards and integrated ESG data platforms provide interactive visualizations and real-time updates, allowing investors to access relevant information more effectively and make informed decisions (Cui, 2025). The shift toward digital disclosure platforms has also been accelerated by structural changes in organizational practices. In particular, the COVID-19 pandemic significantly accelerated digital adoption as remote work environments increased the need for digital-first communication strategies. As organizations adapted to these changes, digital investor reporting systems became an essential component of corporate communication infrastructures (Jaking, 2025). Recent studies also indicate that conventional reporting methods may no longer fully satisfy investor expectations for transparency, sustainability disclosure, and interactive engagement (Adelowotan, 2024).

Despite the growing adoption of digital reporting technologies, academic research has primarily focused on the internal benefits of digital transformation, such as process efficiency and cost reduction. Relatively limited attention has been given to how digital transparency platforms influence investor relations and market valuation (Rydzewski, 2025). This gap is particularly relevant as organizations continue to allocate substantial resources toward digital investor communication systems that may influence investor perceptions and capital market outcomes. Another underexplored issue concerns the relationship between traditional transparency indicators, particularly ESG disclosures, and modern digital transparency infrastructures. As firms simultaneously pursue sustainability objectives and digital transformation strategies, it is important to understand whether these transparency mechanisms complement each other or operate independently in shaping investor perceptions and market valuations (Bednárová, 2025). The intersection of digital transformation and corporate responsibility therefore represents an important area for empirical investigation.

The theoretical foundations for examining digital transparency infrastructure draw from multiple academic disciplines. Information systems research highlights how digital technologies enhance information accessibility and reduce transaction costs (Nauman, 2025). Financial economics suggests that improved information dissemination through digital platforms can enhance market efficiency and reduce firms' cost of capital (Dsouza et al., 2025). In addition, strategic management literature emphasizes digital capabilities as a source of competitive advantage and long-term organizational performance (Geronasso, 2025). However, these theoretical perspectives have rarely been integrated to examine the specific mechanisms through which digital investor reporting systems influence market valuation and stakeholder engagement.

To address this research gap, the present study empirically investigates the impact of data-driven dashboard implementations on investor transparency and market performance among S&P 500 companies. Using cross-sectional data and incorporating both financial indicators and ESG performance

metrics, this study examines how digital transparency infrastructure influences corporate valuation, particularly through Price-to-Earnings (P/E) ratios. Focusing on S&P 500 firms provides both empirical reliability and practical relevance, as these organizations are often leaders in digital adoption and disclosure innovation.

The findings of this research contribute to the literature on digital transformation by providing empirical evidence that investor-oriented digital reporting platforms can influence market valuation through improved disclosure quality and stakeholder engagement. Unlike prior studies that primarily emphasize operational efficiency, this research highlights the external value creation potential of digital transparency technologies (Mölders, 2025). The results have important implications for corporate managers, investors, and regulators, demonstrating that digital reporting infrastructures can serve not only as compliance tools but also as strategic assets for improving investor communication and market perception.

This study therefore examines how data-driven dashboards influence investor reporting transparency in U.S. capital markets. Specifically, it analyzes the relationship between digital reporting quality, ESG performance, and market valuation. The research contributes to existing literature in three ways: first, it provides empirical evidence on the market value implications of digital transparency investments; second, it examines the interaction between digital transformation and traditional ESG disclosure metrics; and third, it presents a sectoral analysis of digital transparency adoption among S&P 500 firms.

The remainder of this paper is structured as follows. Section 2 reviews the relevant literature on digital transformation and ESG transparency. Section 3 describes the research methodology and data analysis procedures. Section 4 presents the empirical findings, while Section 5 discusses their implications. Section 6 outlines the study's limitations and suggests directions for future research. Finally, Section 7 concludes the paper.

2. Literature Review

2.1 Digital Transformation in Financial Services

Digital transformation has become a critical strategic priority across many industries, particularly in the financial sector. Advances in digital technologies have fundamentally changed how financial information is generated, processed, and communicated to stakeholders. Financial institutions and publicly listed companies increasingly rely on digital platforms to enhance operational efficiency and improve communication with investors and regulators.

Research in information systems highlights that digital technologies reduce information asymmetry by improving the accessibility and timeliness of financial data (Brynjolfsson & McAfee, 2014). Similarly, financial economics literature suggests that improved information dissemination enhances market efficiency and lowers firms' cost of capital (Healy & Palepu, 2001). Strategic management research further emphasizes that digital capabilities can serve as a source of competitive advantage, enabling firms to improve decision-making processes and long-term performance (Bharadwaj et al., 2013).

Recent empirical studies also indicate that digital transformation investments contribute to improved financial reporting quality and accounting information comparability. The growing adoption of digital technologies in financial reporting demonstrates that organizations increasingly view digital infrastructure as a strategic asset rather than merely a technical support system. However, much of the existing literature focuses primarily on internal operational benefits, such as automation and cost reduction, while paying relatively little attention to how digital transformation affects investor communication and market valuation.

2.2 Corporate Transparency and Market Performance

Corporate transparency has long been recognized as a key determinant of efficient capital markets. Transparent disclosure reduces information asymmetry between corporate managers and investors, enabling more accurate asset pricing and improving market liquidity (Diamond & Verrecchia, 1991). Traditional measures of corporate transparency typically focus on the timeliness, completeness, and reliability of financial disclosures.

In recent years, transparency frameworks have expanded to include Environmental, Social, and Governance (ESG) reporting. ESG disclosures provide stakeholders with information about a firm's sustainability practices, governance structures, and social responsibilities. Regulatory developments have further strengthened the importance of sustainability reporting. For example, many jurisdictions now require large publicly listed firms to disclose sustainability-related information in their corporate reports.

Previous research suggests that firms with higher transparency attract greater analyst coverage, experience lower information risk, and enjoy improved market valuation (Leuz & Wysocki, 2016). In addition, transparent firms are less likely to engage in earnings manipulation and are more likely to build long-term investor trust (Bushman et al., 2004). Despite the growing relevance of transparency in financial markets, most studies still focus on traditional disclosure mechanisms, while digital transparency platforms remain relatively underexplored.

2.3 Technology-Enhanced Investor Relations

The application of digital technologies in investor relations represents an emerging area of research. Traditionally, investor communication relied on periodic financial reports, earnings conference calls, and investor meetings. However, digital technologies have introduced new forms of communication that allow companies to provide real-time information through interactive platforms.

Digital dashboards, online data portals, and integrated reporting platforms enable firms to present financial and non-financial information in more accessible, interactive formats. These technologies allow investors to visualize complex datasets, conduct real-time analysis, and customize the presentation of financial information according to their analytical needs (Hjelle et al., 2024).

Preliminary research suggests that technology-enhanced investor communication can create value by improving the accessibility and usability of financial data. Digital dashboards reduce information processing costs and provide investors with tools to better interpret corporate disclosures. Despite these advantages, empirical research examining the impact of digital reporting technologies on market valuation remains limited. In addition, the relationship between digital transparency systems and traditional disclosure mechanisms such as ESG reporting has not been extensively investigated.

2.4 Research Gap and Theoretical Framework

The existing literature reveals several important research gaps. First, empirical evidence evaluating the market value implications of digital transparency investments remains limited. While previous studies have examined digital transformation in operational contexts, relatively little research has focused on digital reporting technologies designed specifically for investor communication. Second, the interaction between traditional transparency indicators, particularly ESG performance, and digital transparency infrastructure has not been adequately explored. As organizations simultaneously adopt sustainability reporting frameworks and digital reporting platforms, understanding the relationship between these transparency mechanisms is increasingly important. Third, existing research provides limited insights into sectoral differences in the adoption and effectiveness of digital reporting technologies. Industry-specific characteristics, regulatory pressures, and technological capabilities may influence how firms

implement digital transparency initiatives. To address these gaps, this study adopts a theoretical framework grounded in information asymmetry theory and signalling theory. From an information asymmetry perspective, digital dashboards reduce informational gaps between firms and investors by providing timely and accessible financial information. From a signalling theory perspective, investments in advanced digital transparency infrastructure signal managerial competence and organizational capability, potentially enhancing investor confidence and market valuation. Based on this framework, the study hypothesizes that firms with more advanced digital transparency infrastructure are likely to experience higher market valuation and improved investor perception.

3. Methodology

3.1 Research Design

This study employs a quantitative cross-sectional research design to examine the relationship between digital transparency infrastructure and market performance among publicly listed firms. The empirical analysis investigates how the implementation and quality of digital investor reporting systems influence corporate valuation. A combination of descriptive statistics, correlation analysis, and multiple regression models is used to evaluate the relationships between digital reporting quality, ESG performance, and market outcomes. The S&P 500 Stock Data provides historical price information enabling volatility and return calculations, all of which were downloaded from Kaggle (<https://www.kaggle.com/datasets/rikinzala/s-and-p-500-esg-and-stocks-data-2023-24>) and (<https://www.kaggle.com/datasets/ilyaryabov/financial-performance-of-companies-from-sp500/data>). Our sample includes 47 companies representing eight sectors: Communication Services, Consumer Discretionary, Consumer Staples, Energy, Financials, Healthcare, Industrials, and Technology. This cross-sectional design allows for a comprehensive analysis of digital transparency implementations across diverse industry contexts. Company selection ensured representative coverage of market capitalizations and growth profiles. The sample includes both established large-cap companies and high-growth technology firms, providing generalizability to broader S&P 500 populations. For more clarity, the schematic representation is diagram 1.

3.2 Data Sources and Sample Selection

The dataset used in this study was compiled from publicly available financial and ESG databases obtained from Kaggle. Two primary datasets were utilized:

- **S&P 500 Financial Performance Dataset**, which includes financial indicators such as revenue, income, market capitalization, and valuation metrics.
- **S&P 500 ESG Dataset**, which provides environmental, social, and governance performance scores for listed companies.

After merging and cleaning the datasets, the final sample consists of 47 S&P 500 companies across eight industry sectors: Communication Services, Consumer Discretionary, Consumer Staples, Energy, Financials, Healthcare, Industrials, and Technology. The sample includes firms with varying levels of market capitalization and financial performance, enabling a comprehensive analysis of digital transparency adoption across different industries.

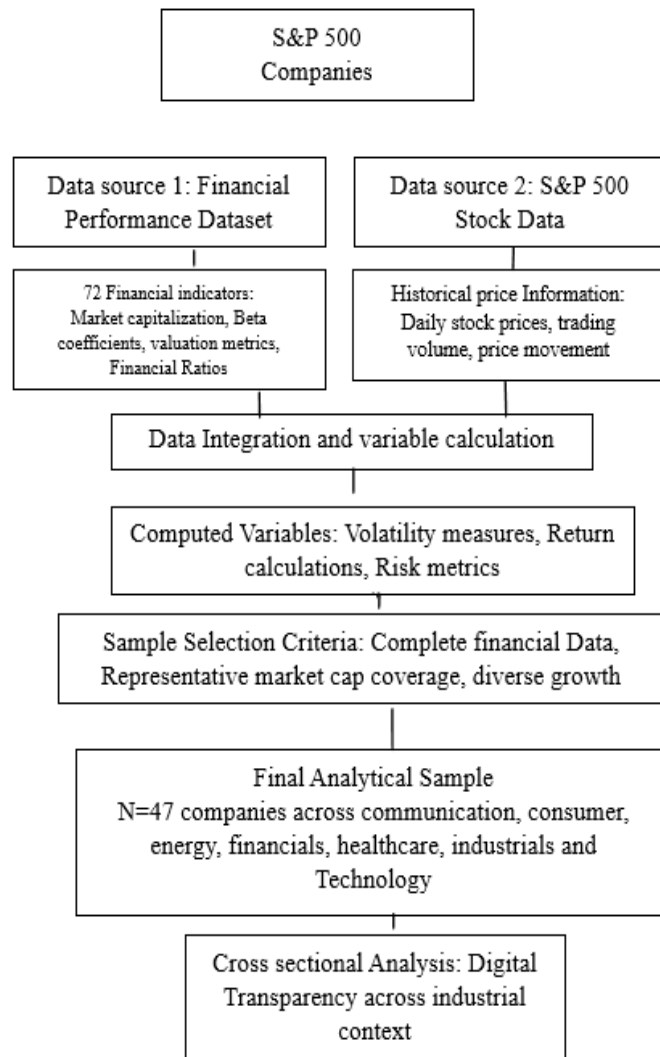


Diagram 1

3.3 Variable Definitions

3.3.1 Dependent Variables

The study measures market performance using three indicators:

- **Price-to-Earnings Ratio (P/E TTM)** – primary measure of market valuation
- **Forward Price-to-Earnings Ratio** – indicator of expected growth
- **Analysts’ Mean Recommendation** – proxy for investor sentiment

3.3.2 Independent Variables

The main explanatory variables represent digital transparency infrastructure:

- **Uses Dashboard** – binary variable indicating whether a company uses a data-driven reporting dashboard
- **Quality Score** – ordinal measure (scale 3–5) reflecting the sophistication of digital reporting systems
- **Total ESG Score** – composite sustainability performance indicator
- **Environmental, Social, and Governance scores** – individual ESG components

3.3.3 Control Variables

Several control variables are included to account for firm-level differences:

- **Market Capitalization** – proxy for company size
- **Beta** – systematic risk measure
- **Revenue (TTM)** – operational scale indicator
- **Income (TTM)** – profitability indicator
- **Sector** – industry classification

3.4 Analytical Methods

3.4.1 Descriptive Analysis

Descriptive statistics were first computed to summarize the characteristics of the dataset, including mean values, standard deviations, and distribution ranges of the key variables.

3.4.2 Comparative Analysis

To examine differences between firms with and without digital dashboards, independent sample t-tests and Mann–Whitney U tests were conducted. These tests assess whether statistically significant differences exist between the two groups in financial performance and ESG scores.

3.4.3 Regression Analysis

To assess the impact of digital transparency infrastructure on market valuation, the following regression model was estimated:

Market Performance

$$= \beta_0 + \beta_1 \text{Dashboard} + \beta_2 \text{Quality} + \beta_3 \text{ESG} + \beta_4 (\text{Quality} \times \text{ESG}) + \beta_5 \text{Controls} + \varepsilon$$

This model evaluates both the direct effects of digital reporting quality and the interaction between digital transparency and ESG performance.

3.4.4 Mediation Analysis

To further examine the relationship between ESG performance and market outcomes, mediation analysis based on the **Baron and Kenny (1986)** framework was conducted. This approach evaluates whether digital reporting quality mediates the relationship between ESG performance and firm valuation.

3.4.5 Robustness Checks

Additional robustness tests were conducted to ensure the reliability of the results. These include:

- alternative model specifications
- outlier sensitivity analysis
- sector-specific regression models

These procedures help confirm the stability and validity of the empirical findings.

4. Results

4.1 Descriptive Statistics

This study analyzes 47 S&P 500 companies across eight industry sectors to evaluate the relationship between digital transparency infrastructure and market performance. Table 1 presents descriptive statistics for the main variables. The total ESG score ranges from 11.78 to 41.60, with a mean value of 23.80 and a standard deviation of 8.15, indicating moderate variation in sustainability performance among firms. The Quality Score, which measures the sophistication of digital reporting systems, ranges from 3 to 5, with a mean of 3.93. Market valuation indicators also display significant variation across the sample. The price-to-earnings (P/E) ratio ranges from 6.48 to 888.11, with an average value of 49.78, reflecting differences in investor expectations and growth prospects across firms. Revenue and income variables also demonstrate wide dispersion, reflecting differences in firm size and operational scale. Overall, the descriptive results indicate substantial variation in both digital reporting quality and financial performance across the sampled firms. The distribution of price-to-earnings ratios between firms with and without dashboard usage is presented in Figure 1.

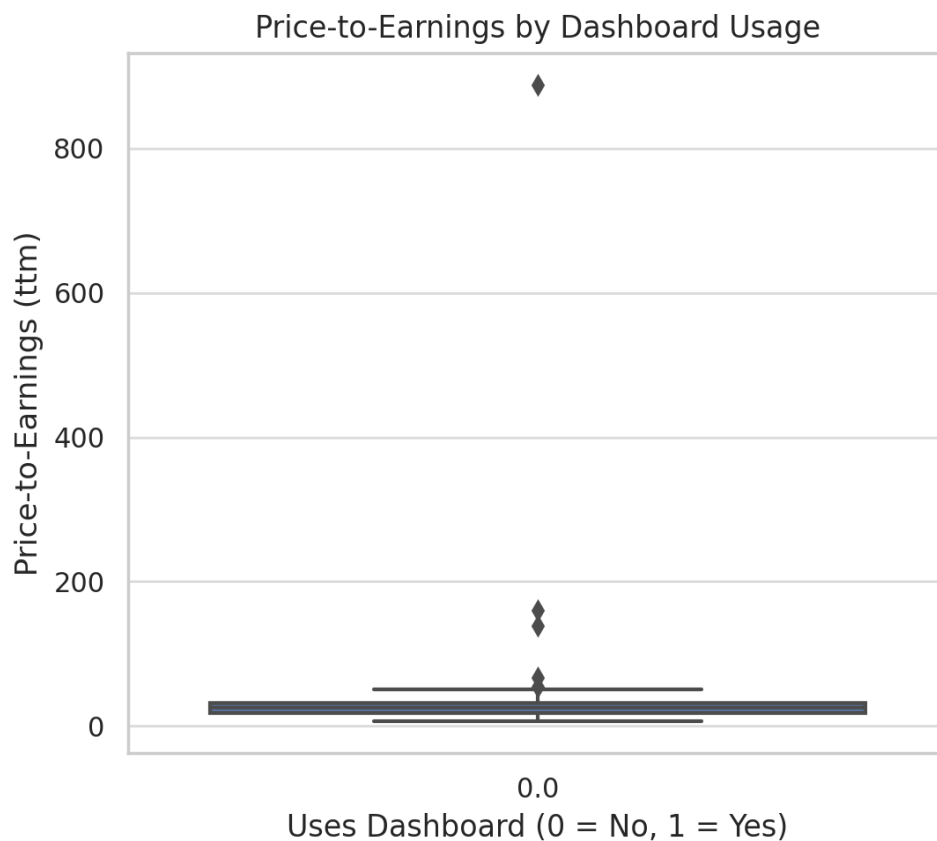


Figure 1: The boxplot displays the distribution of Price-to-Earnings (P/E) ratios based on Dashboard Usage (0 = No, 1 = Yes). The plot shows the median, interquartile range, and any outliers for both groups. The plot highlights that most companies, regardless of dashboard usage, have relatively low P/E ratios, with only a few companies exhibiting extreme values (outliers).

Table 1: Descriptive Statistics of Key Financial and ESG Variables

Metrics	count	mean	std	min	25%	50%	75%	max
Environment Score	47	4.787234	5.191212	0.03	1.45	2.3	7.415	23.11
Social Score	47	11.16	4.09436	4.89	7.93	10.64	14.16	22.48

Governance Score	47	7.853617	2.665187	3.17	6.255	7.19	9.06	19.43
Total Esg	47	23.79851	8.145381	11.78	16.875	23.98	28.595	41.6
Price-to-Earnings (ttm)	45	49.77533	131.0382	6.48	17.78	25.32	31.8	888.11
Forward Price-to-Earnings	47	18.45574	8.382448	6.25	12.1	17.14	22.34	41.22
Income (ttm)	47	1.52E+10	1.93E+10	-4.9E+09	5.5E+09	8.91E+09	1.6E+10	9.52E+10
Revenue (ttm)	47	1.12E+11	1.13E+11	1.76E+10	4.61E+10	6.87E+10	1.08E+11	5.14E+11
Beta	47	1.023915	0.424518	0.399	0.7075	1.027	1.2205	2.29
Quality Score	47	3.925532	0.466004	3	3.5	4	4	5

4.2 Sectoral Analysis

Sectoral comparisons reveal differences in both ESG performance and digital reporting sophistication. As shown in Table 2, the Energy sector exhibits the highest average ESG score (37.37), followed by Industrials (35.02). These results likely reflect stronger regulatory and stakeholder pressures related to sustainability reporting in these industries. In contrast, the Technology sector demonstrates the highest digital reporting quality (Quality Score = 4.40) despite having relatively lower ESG scores (15.45). This finding suggests that firms with strong technological capabilities may prioritize advanced digital disclosure platforms over traditional sustainability reporting indicators. The Communication Services sector shows a balanced profile, combining relatively high ESG performance with advanced digital reporting systems. These sectoral patterns highlight differences in transparency strategies across industries. The relationship between ESG performance and market valuation is illustrated in Figure 2, which shows the distribution of Price-to-Earnings ratios across firms.

Table 2. ESG and Financial Performance by Sector

Sector	Total Esg	Price-to-Earnings (ttm)	Forward Price-to-Earnings	Quality Score
Communication Services	24.27	31.98	14.56	4.25
Consumer Discretionary	22.44	29.96	24.19	3.79
Consumer Staples	24.61	28.04	21.57	3.38
Energy	37.37	8.20	10.58	4.00
Financials	26.83	14.07	10.55	3.83
Healthcare	20.95	22.04	14.02	3.88
Industrials	35.02	57.21	22.75	3.50
Technology	15.45	129.00	21.90	4.40

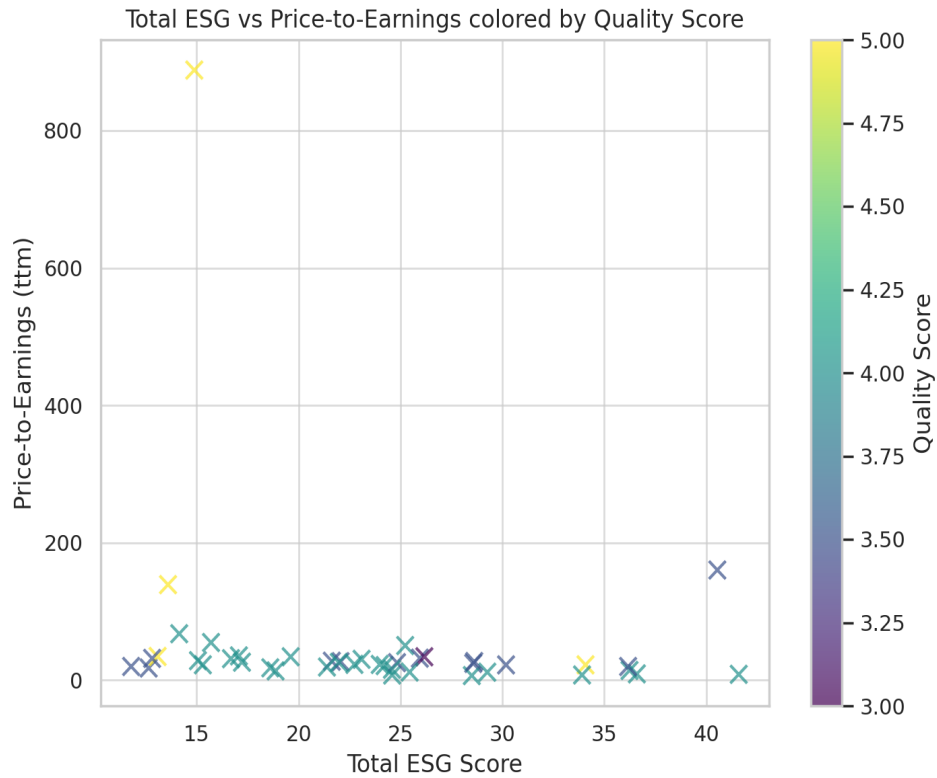


Figure 2: The scatter plot displays the relationship between Total ESG Score and Price-to-Earnings (P/E) ratio for the selected companies. The points are coloured according to the Quality Score, with warmer colors representing higher Quality Scores. This plot illustrates the association between ESG performance and investor reporting quality (as measured by the Quality Score) and stock performance (measured by the P/E ratio).

4.3 Correlation Analysis

Table 3 and Figure 3, the correlation matrix, displays several noteworthy linkages that support our theoretical framework. The total ESG scores are very strongly associated with the individual ESG components: Environment ($r = 0.70$), Social ($r = 0.73$), and Governance ($r = 0.57$). This suggests that our composite measure is correct. Quality Score is only weakly related to Total ESG ($r = -0.22$). This means that companies with better digital reporting systems might care more about different parts of openness. On the other side, Quality Score shows a strong positive relationship with Beta ($r = 0.41$) and Price-to-Earnings ratios ($r = 0.41$). This means that more complex dashboard setups are associated with more valuable equities that are more likely to fluctuate in value. The relationship between ESG performance and firm income is visualized in Figure 4 and Figure 5.

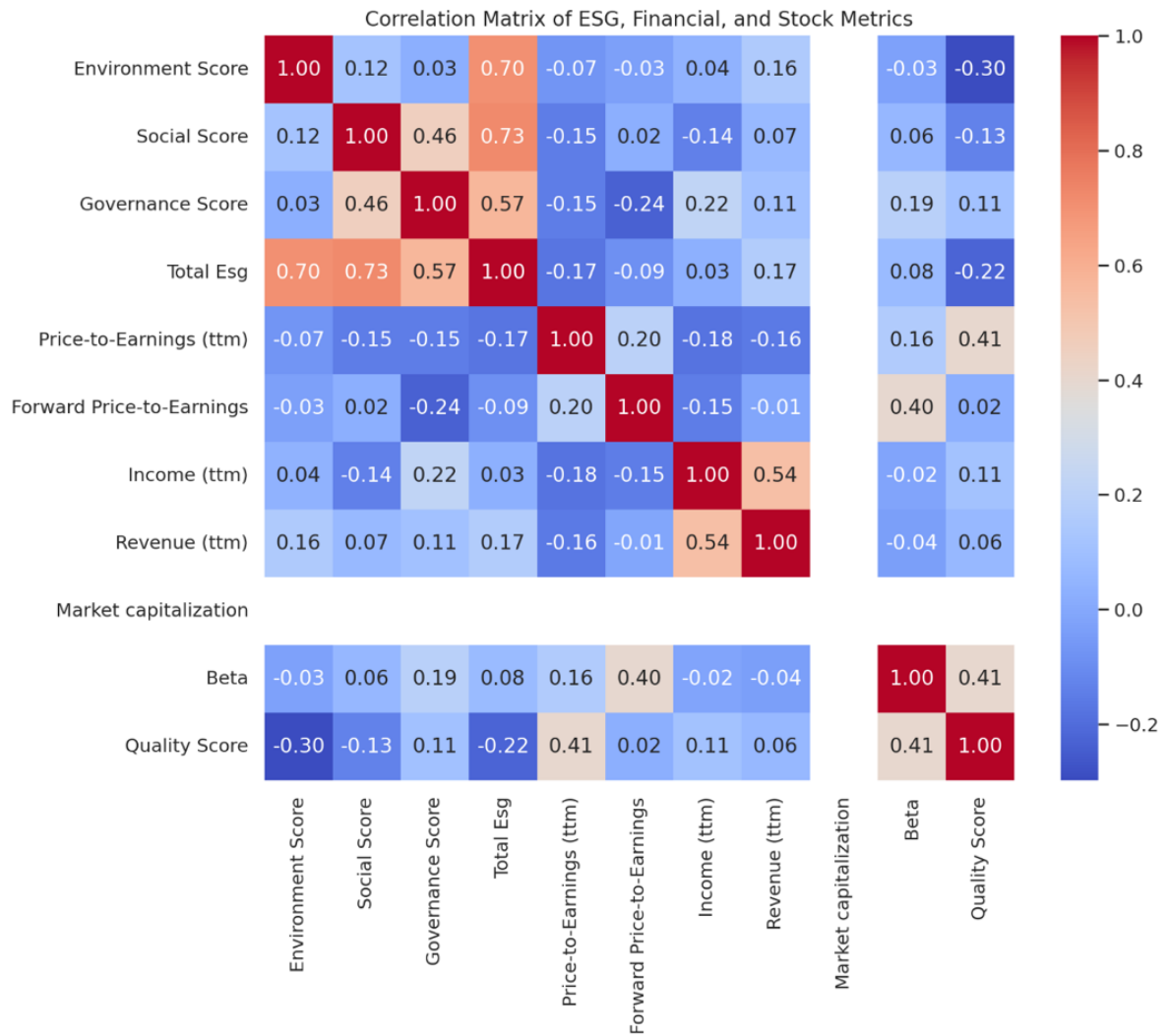


Figure 3: This heatmap presents the Pearson correlation matrix for key ESG scores, financial metrics, and stock performance indicators. The colours represent the strength of the correlations, with warmer colours indicating stronger positive correlations and cooler colours indicating negative correlations. The matrix reveals significant positive correlations among Total ESG and its components, as well as between Income (ttm) and Revenue (ttm). Notably, Quality Score has a moderate negative correlation with Environment Score and a positive correlation with Price-to-Earnings (P/E).

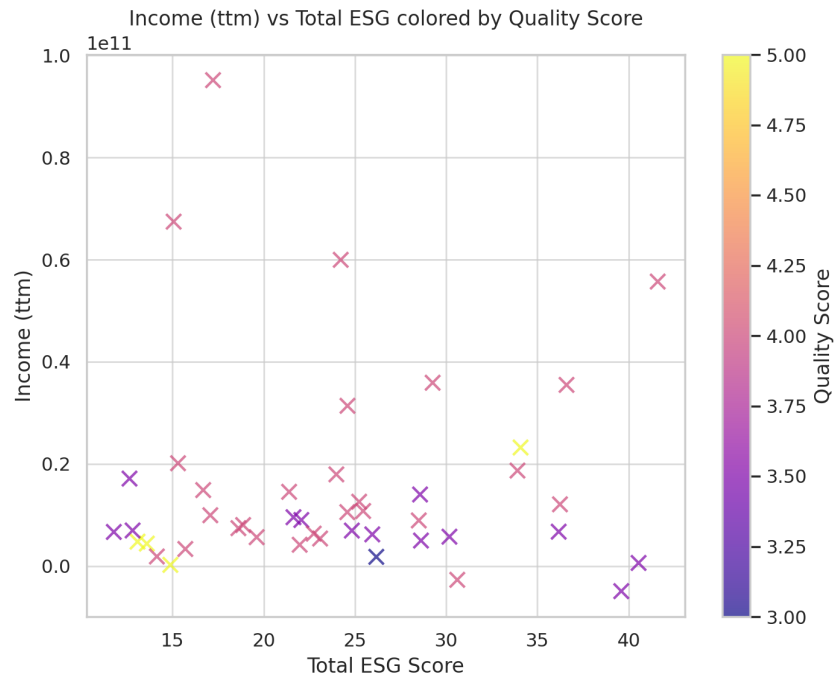


Figure 4: The scatter plot illustrates the relationship between Total ESG Score and Income (ttm), with points coloured according to the Quality Score. The colour gradient shows higher Quality Scores in warmer colours, indicating better-quality reporting. The plot suggests that companies with higher Total ESG Scores tend to have higher Income (ttm), especially those with higher Quality Scores.

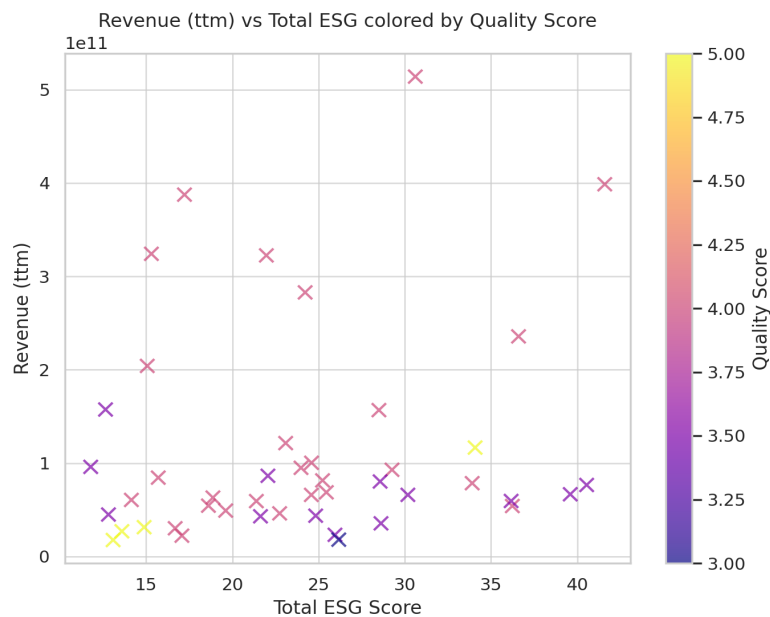


Figure 5: The scatter plot illustrates the relationship between Total ESG Score and Revenue (ttm), with points colored based on the Quality Score of each company. The color gradient reflects the range of Quality Scores, with warmer colors indicating companies with higher reporting quality. The plot indicates a positive trend where companies with higher Total ESG Scores generally exhibit higher Revenue (ttm), particularly those with higher Quality Scores.

Table 3: Correlation Matrix of ESG Scores, Financial Metrics, and Stock Performance Indicators

Metrics	Environment Score	Social Score	Governance Score	Total Esg	Price-to-Earnings (ttm)	Forward Price-to-Earnings	Income (ttm)	Revenue (ttm)	Beta	Quality Score
Environment Score	1	0.115170876	0.025328366	0.703343905	-0.073031904	-0.033949438	0.037073786	0.155695653	-0.02824	-0.298071495
Social Score	0.115170876	1	0.456254613	0.725193246	-0.150730369	0.022608827	-0.138052641	0.067277729	0.060448	-0.13336992
Governance Score	0.025328366	0.456254613	1	0.572703074	-0.149605849	-0.236767967	0.223578766	0.105240999	0.192249	0.107173686
Total Esg	0.703343905	0.725193246	0.572703074	1	-0.171745853	-0.087876594	0.027368945	0.167620869	0.075023	-0.223454538
Price-to-Earnings (ttm)	-0.073031904	-0.150730369	-0.149605849	-0.171745853	1	0.203844807	-0.178398725	-0.158066452	0.160514	0.405687235
Forward Price-to-Earnings	-0.033949438	0.022608827	-0.236767967	-0.087876594	0.203844807	1	-0.146242237	-0.010159482	0.400494	0.024106026
Income (ttm)	0.037073786	-0.138052641	0.223578766	0.027368945	-0.178398725	-0.146242237	1	0.537864901	-0.01767	0.108489303
Revenue (ttm)	0.155695653	0.067277729	0.105240999	0.167620869	-0.158066452	-0.010159482	0.537864901	1	-0.03885	0.062823734
Beta	-0.028243184	0.060448243	0.192248558	0.075023421	0.16051437	0.400493858	-0.017667955	-0.0388538	1	0.406403987
Quality Score	-0.298071495	-0.13336992	0.107173686	-0.223454538	0.405687235	0.024106026	0.108489303	0.062823734	0.406404	1

4.4 Regression Analysis: Dashboard Impact on Market Performance

Table 4 presents the results of our main regression model, which examines how digital transparency initiatives affect stock market performance, as indicated by Price-to-Earnings ratios. The model attained statistical significance and elucidated numerous critical findings:

H1: The Digital Dashboard Quality and Market Valuation: Quality Score have a substantial positive relationship ($\beta = 402.97$, $p < 0.01$), which means that organizations with more advanced data-driven dashboards are worth a lot more on the market. This indicates that investors appreciate improved transparency infrastructure, corroborating our core premise that digital transformation is important for mitigating information asymmetries.

H2: Synergies between ESG Performance and Transparency: There is a positive main impact for total ESG scores ($\beta = 49.92$, $p < 0.05$), but this effect is weakened by Quality Score, which has a significant negative interaction term ($\beta = -13.08$, $p < 0.05$). This interaction indicates that the market premium for ESG performance is greatest among companies with moderate digital reporting capabilities, but more advanced digital platforms may diminish the additional valuation advantage of conventional ESG measurements.

Table 4: Regression Coefficients for the Model with Total ESG, Quality Score, and Beta

Variable	Coefficient	P-value
Intercept	-1502.49928	0.002338
Q("Total Esg")	49.91648873	0.014006
Q("Quality Score")	402.9676768	0.00132
Q("Total Esg"):Q("Quality Score")	-13.0834755	0.010637
Q("Beta")	-4.28007331	0.936015

4.5 Mediation Analysis

Table 5 presents the mediation analysis examining how Quality Score affects the link between Total ESG performance and market outcomes. The analysis shows a complex mediation pattern, with both direct ($\beta = -1.76$) and indirect ($\beta = -1.39$) effects summing to the total effect ($\beta = -3.15$). This pattern indicates that both ESG performance and digital reporting quality independently affect market outcomes; nevertheless, their interplay generates intricate dynamics in which conventional ESG criteria may be undervalued in the presence of superior digital transparency platforms. Investor sentiment regarding ESG performance and reporting quality is illustrated in Figure 6.

Table 5: Direct and Indirect mediation analysis of Total ESG on Stock Performance via Quality Score

Effect Type	Effect Value
Indirect (via Quality Score)	-1.393882916
Direct (Total ESG)	-1.762220399
Total Effect	-3.156103315

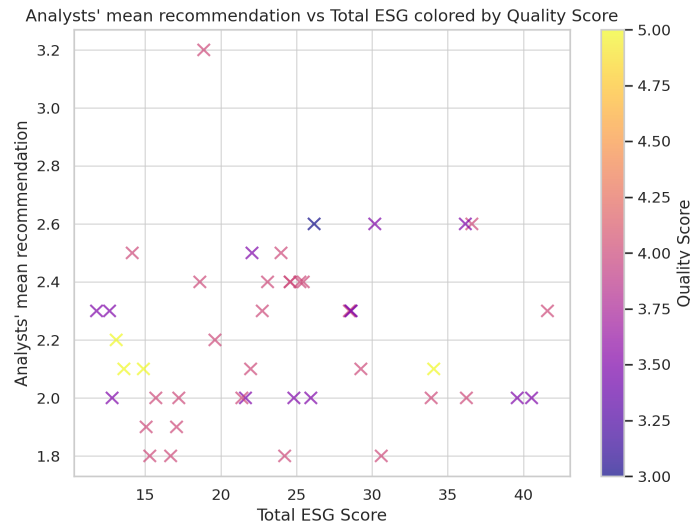


Figure 6: The scatter plot illustrates the relationship between Total ESG Score and Analysts' Mean Recommendation, with points coloured according to the Quality Score. The gradient colour scale represents the Quality Score, with warmer colours indicating higher-quality reporting. The plot shows that companies with higher Total ESG Scores tend to receive better Analyst Recommendations, especially those with higher Quality Scores, suggesting that transparent reporting influences investor sentiment.

5. Discussion

The findings of this study improve our understanding of how digital transformation influences capital market mechanisms and stakeholder value creation. In particular, the results contribute to the digital transformation, information economics, and corporate finance literature by providing empirical evidence that extends existing theoretical frameworks (Zhang and Wang, 2024; Westerman, 2014).

The strong positive relationship between digital reporting quality and market valuation ($\beta = 402.97$, $p < 0.01$) supports the argument that digital transformation can reduce information asymmetry between firms and investors (Chen and Srinivasan, 2024). These findings reinforce information economics theory by demonstrating that technological solutions, such as digital dashboards, enhance transparency and reduce information-processing costs for market participants (Lin, 2016). Unlike traditional reporting mechanisms that rely on periodic disclosure formats, digital dashboards provide continuous, customizable, and interactive access to information, fundamentally changing how firms communicate with investors (Sousa-Zomer et al., 2020). The results also suggest that improvements in digital reporting quality significantly enhance firm valuation. Investors appear to place considerable value on accessible and high-quality information environments, indicating that the effectiveness of information distribution systems contributes to market valuation beyond the content of the disclosures themselves (Basoglu and Hess, 2014). These findings further support signalling theory, suggesting that advanced digital transparency infrastructure signals managerial capability, operational efficiency, and strategic competence to investors (Saleem, 2022). Firms that invest in digital transparency platforms may therefore attract more sophisticated investors, increase analyst coverage, and ultimately improve their market valuation (Westerman, 2014).

However, the negative interaction between ESG performance and digital reporting quality ($\beta = -13.08$, $p < 0.05$) suggests that traditional and digital transparency mechanisms may not always operate as complementary factors (Basoglu and Hess, 2014). Instead, the findings indicate a substitution effect, where advanced digital reporting systems may reduce the marginal valuation impact of traditional ESG indicators. As digital transparency improves, investor expectations may shift toward real-time information accessibility and analytical capability rather than relying solely on historical sustainability disclosures (Chen and Srinivasan, 2024).

This interaction effect highlights the possibility of diminishing returns in transparency strategies. Firms with highly advanced digital reporting infrastructure may benefit less from incremental improvements in ESG reporting, suggesting that optimal transparency strategies require a balanced combination of traditional and digital disclosure mechanisms (Lin, 2016). These results are consistent with the resource-based view, which suggests that certain organizational capabilities may substitute for others rather than operate purely as complementary resources (Zhang and Wang, 2024).

The sectoral analysis further illustrates how industry context influences the adoption of digital transparency. Technology firms exhibit the highest digital reporting quality (Quality Score: 4.40) despite relatively lower ESG scores, while sectors such as Energy demonstrate stronger ESG performance but slightly lower digital reporting sophistication (Sousa-Zomer et al., 2020). These patterns suggest that sector-specific regulatory pressures, technological capabilities, and stakeholder expectations shape corporate transparency strategies. Technology firms benefit from stronger digital competencies and innovation-oriented cultures, whereas industries such as Energy and Industrials face stronger regulatory scrutiny related to environmental sustainability (Saleem, 2022). From an institutional perspective, these sectoral differences reflect how regulatory environments and stakeholder pressures influence corporate disclosure practices. Firms operating in environmentally sensitive sectors face stronger external demands for sustainability reporting, while technology firms often compete through digital innovation and technological capabilities (Westerman, 2014). As a result, transparency strategies may evolve differently across industries depending on institutional pressures and competitive dynamics.

The findings also have important managerial implications. The strong association between digital reporting quality and market valuation provides a clear business case for firms to invest in digital transparency infrastructure (Chen and Srinivasan, 2024). As relatively few organizations report achieving substantial success in digital transformation initiatives, firms that successfully implement advanced digital reporting systems may gain a significant competitive advantage (Zhang and Wang, 2024). These investments should therefore be viewed as strategic initiatives that generate measurable financial returns rather than merely operational improvements. The interaction between ESG performance and digital reporting quality also provides guidance for corporate disclosure strategies. Firms with moderate digital capabilities may continue to benefit from strengthening traditional ESG performance, while firms with advanced digital transparency systems may achieve greater returns by prioritizing technological sophistication and real-time information accessibility (Sousa-Zomer et al., 2020; Westerman, 2014). Understanding these substitution effects allows firms to allocate resources more effectively between sustainability initiatives and digital transparency investments (Lin, 2016).

The results also offer insights for investors. The positive relationship between digital reporting quality and analyst recommendations suggests that investors recognize the value of advanced digital reporting systems when evaluating firms (Saleem, 2022). Consequently, investors may benefit from incorporating assessments of digital transparency infrastructure into their investment analysis and due diligence processes (Zhang and Wang, 2024). Finally, the findings have implications for regulators and policymakers. Evidence that digital transparency infrastructure influences market valuation suggests that regulatory frameworks may need to evolve to accommodate emerging digital reporting technologies (Sousa-Zomer et al., 2020). Regulatory standards may need to address issues such as data accessibility, reporting standardization, and digital disclosure quality in order to ensure that technological advancements enhance rather than undermine investor protection (Lin, 2016; Westerman, 2014). Overall, this study highlights the growing importance of digital transparency infrastructure in capital markets and suggests that the interaction between digital reporting systems and traditional transparency mechanisms represents an important area for future research (Chen and Srinivasan, 2024).

6. Limitations and Future Work

While this study offers important insights into how digital transparency affects market performance, it has a few limitations. First, the study relies on secondary data from a single year (2023), which limits

our ability to fully capture the long-term impact of digital transformation. Second, focusing solely on S&P 500 companies may limit the broader applicability of the findings, especially for smaller or non-public companies. Future research could benefit from examining data over a longer period to see how digital transparency evolves. It would also be useful to include data from other stock indices or markets outside the U.S. to see if these findings hold in different contexts. Finally, it would be exciting to explore how emerging technologies like AI and blockchain could further enhance digital transparency in investor reporting, offering fresh perspectives on this growing field.

7. Conclusion

Data-driven dashboard deployments greatly impact investor views and market valuations in American capital markets, according to this study. Companies with advanced digital transparency infrastructure earn market premiums. We found a 402.97 ($p < 0.01$) coefficient for Quality Score in predicting Price-to-Earnings ratios. The study adds to digital transformation literature by showing that investor engagement technologies can pay off. Our research, unlike operational efficiency studies, shows how transparency technologies can create market value. This discovery supports business digital transformation beyond internal operations improvements. The findings affect several stakeholders. Given high market premiums, business executives should invest heavily in digital reporting infrastructure. Investors should examine a company's transparency and technological efficiency when assessing its quality. Policymakers must analyse how the digital revolution affects transparency and market efficiency. Future research might examine digital transformation's effects over time, increase transparency metrics for digital infrastructure, and examine how technological skills create market value. The rapid pace of technological innovation and changing legislation presents scholars with several new research opportunities. Understanding technology and transparency is crucial as financial markets digitise. This research suggests that this convergence creates value, boosting digital transformation investments and underlining the changing nature of corporate openness in modern marketplaces.

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Conflicts of Interest

The author(s) have no conflict of interest to declare.

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